

Lab 1

Introduction

* Genotyping

→ mutation $< 1\%$

→ SNP (Single nucleotide polymorphism) $\geq 1\%$

mutation ← heterozygous / homozygous mutant

variant
Solid mutation / variant

SNP ← homovariant

* Genotyping steps:

1 Nucleic Acids

Isolation

extraction
purification

DNA

RNA

plasmid

from

E. coli

blood cell

blood cell

bacteria

using organic
methods

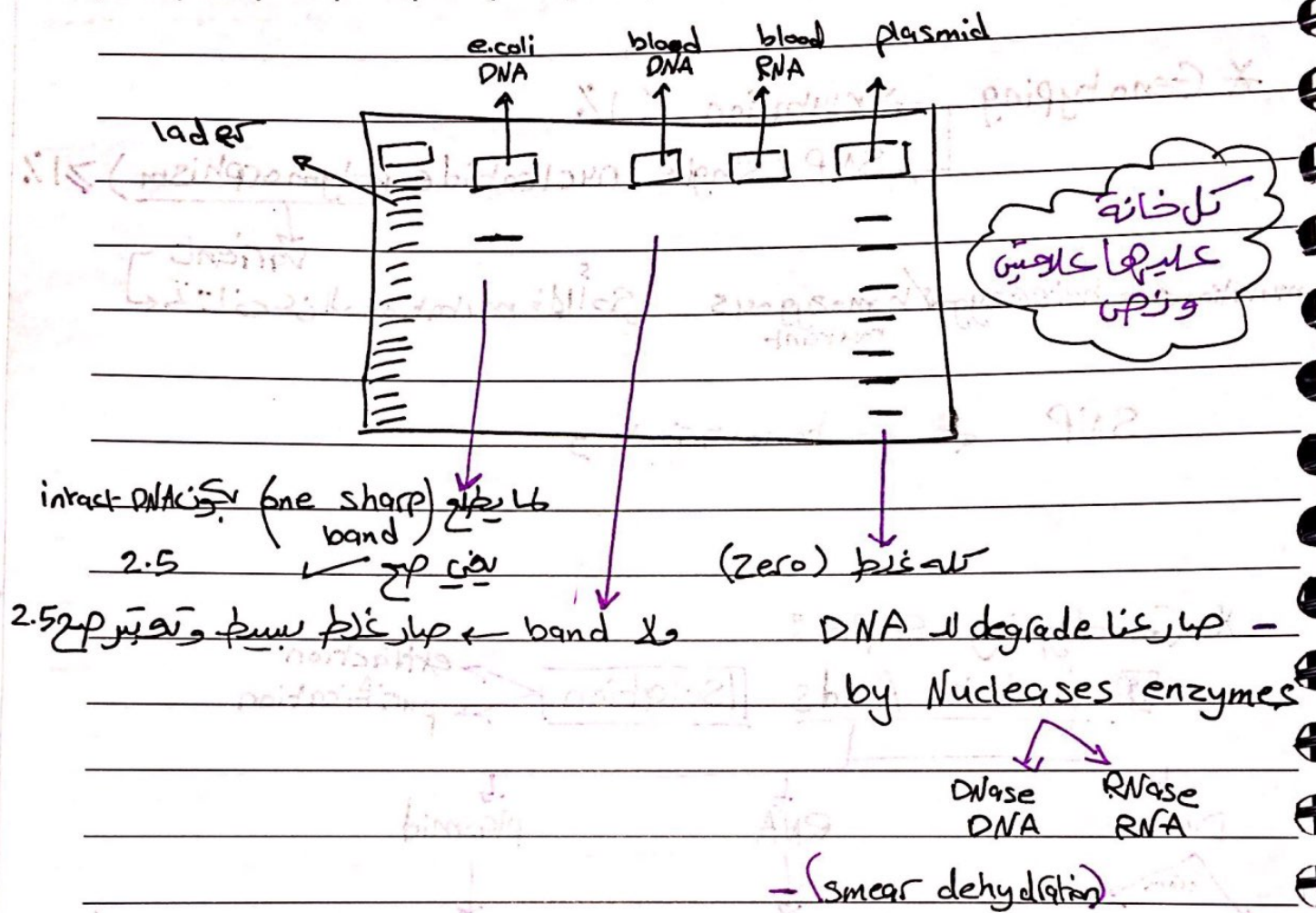
salting
out

adsorption
column
(spin)

2 Nucleic acids analysis & Quantitative

- Agarose gel electrophoresis

• semiquantitative method gives you indication.



2kbs 3 PCR

4 Genotyping: (DNA extracted used as a template in molecular techniques)

1) Allele specific PCR

2) RFLP (restriction Fragment length polymorphism)

3) sequencing

4) Cloning (for plasmid)

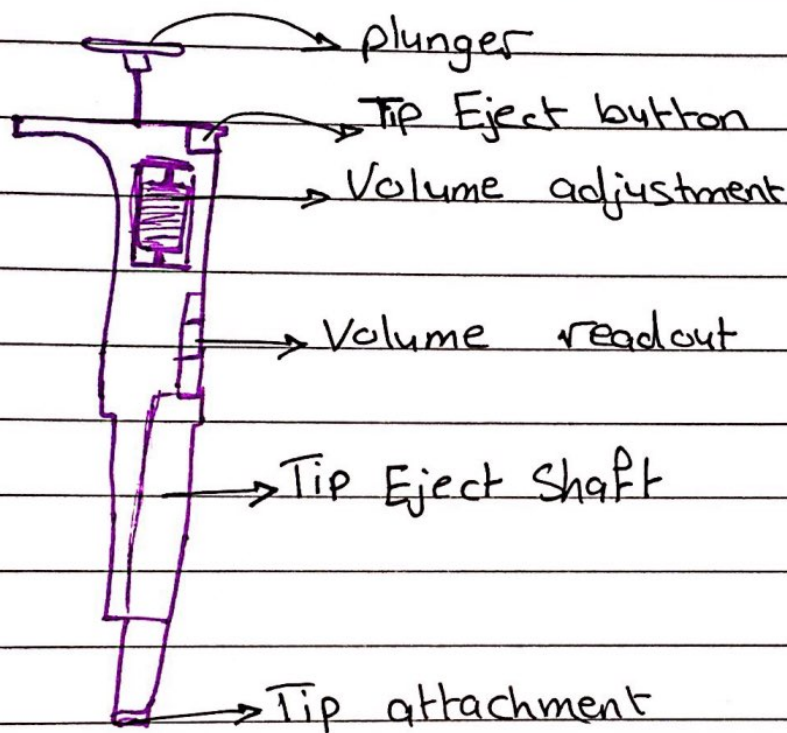
Data base lab: قبل بان PCR (reverse transcription) RNA الى C-DNA بواسطة (reverse transcriptase enzyme)

RT-PCR

5) Southern Blotting.

* Micropipette $\Rightarrow$ Calibrated glass tube into which the ~~liquid~~ liquid is drawn and then released

Func $\rightarrow$ to accurately and precisely deliver small volumes of liquid.



* احسان بن علي
dc / ab / Mate
950127

1 Blue

100-1000 μ l

2 yellow

20-200 μ l

3 Red

0.5-10 μ l

350

0	thousand
3	hundred
5	tens

75

0	hundred
7	tens
5	units

10.5

1	tens
0	units
5	tenth

* Tips are loaded into tips box and sterilized to prevent contamination. (don't touch it by your finger)

* DNA isolation from E. coli (bacteria)

using organic method

Overnight culture of E. coliPelletlysispartitioning of DNA & proteinsDNA precipitationDNA washingDNA rehydration & storageMajor steps in isolation :

- 1) Cell lysis
- 2) Removal of contaminants
- 3) DNA precipitation
- 4) DNA washing
- 5) DNA rehydration

* To isolate DNA, you must remove :

- 1) cell membrane
- 2) cell wall
plant, bacteria, fungi
- 3) cytoplasm
- 4) Nuclear membrane
animal, plant
- 5) proteins, polysaccharide & lipids.

* methods of Isolation:

- Differential Solubility: organic solvents

- Salting out

- Adsorption methods

- Density gradient centrifugation

[1] Bacteria pelleting & resuspension

TE: Tris-EDTA

→ EDTA: binds up Mg^{++} ions

(necessary cofactor of DNases)

EDTA: Ethylene Diamine
tetra acetic acid

(chelating agent)

that remove divalent
ions (+2) from solution

← and sequestering these ions and
prevent their interference.

→ TRIS: maintain pH of DNA
 ≈ 8

[2] Bacteria lysis

→ NaCl: stabilise DNA and remains as
double helix

→ SDS: Sodium dodecyl sulfate
- change folding and conformations
to protein leads to denature it

DNA depoteinization
and inactivation of

- Surfactant (hydrophobic) $\rightarrow$ H_2O
(hydrophilic) $\rightarrow$ proteins

DNase elevated temp

- It is negative charge (Anionic) binds
with positive charge of protein (histone)
to release DNA.

→ Proteinase K: For sureness to denature
all contaminated proteins
(keratin (hair protein))

[3] Organic solvent extraction:

→ Phenol: removes protein

→ chloroform and isoamyl alcohol

dissolve carbohydrate

lipids & phenol

AntiFoaming

تسنع الرغوة



[4] DNA precipitation

→ Sodium Salt:

ال DNA يتكون من روابط H_2O و Na^+ تستخدم Na^+ لتسريع
تسريع ال H_2O وتربط مع ال Na^+ لأنها اقوى
من (P-H).

after binding $\rightarrow Na^+$ decrease DNA
negative charge allowing it to come
closer and spool. $[NaCl] \rightarrow$ to keep
DNA in its double helix form.

→ Isopropanol / ethanol:

affinity between (Isopro - H_2O) >
(Isopro - DNA) So Isopro take H_2O
and precipitate DNA.

[5] DNA washing

→ 70% ethanol: remove residual
salts and other contaminants.

(dissolve Na^+ and isopro-)

[6] Air drying → removes ethanol

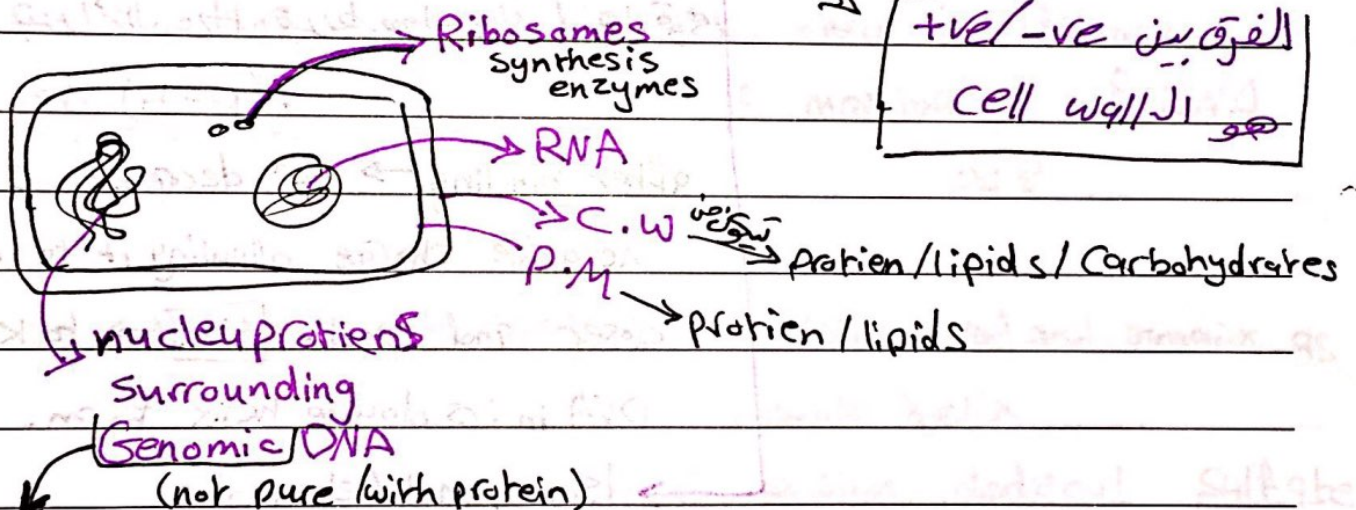
[7] resuspension → H₂O or TE: dissolve DNA pellet

* DNA isolation from E. scherichia coli:

why?

It's nonpathogenic

→ E. coli is gram -ve bacteria



عشان نفرق
بينها وبين
ال plasmid

* Steps for procedure:

← اول خطوة هي ال protection لل DNA عن طرقة nuclease inactiv.

(عشان نفتح opening عشان نطلع ال DNA ونجبر ال Denaturation)

خطوة ٢ : microcentrifuge tube = eppendorf

وسعة تتراوح بين 0.2-2 ml

خطوة ٣ : $\Rightarrow$ Decant = discard : نقي السائل
وبنحلي pellet

هنا الخطوة اسمها (Harvesting)

خطوة ٤ : resuspend by TE buffer

يتم إعادة تعليقها في TE buffer

في أنبوب DNAse إلى نزع degradation للمكونات

إلى الداخل و re suspension

منزوعة بال eppendorf

T.E $\rightarrow$ EDIA

$\rightarrow$ Tris (شعاع قبل)
بالتفصيل

خطوة ٥ : proteinase K / SDS (مضروبة قبل)

incubate at 37°C $\rightarrow$ optimum temp for this enzyme

1 hr

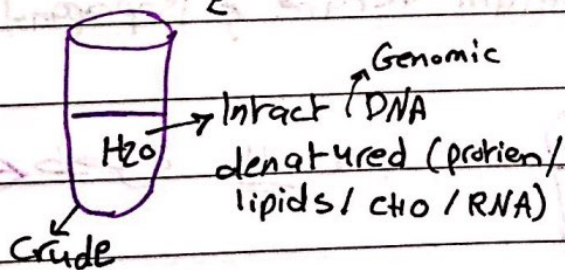
* SDS هو كسر الـ p.m/c

وال histone-like

nucleoproteins

proteinase K *

كسر peptide bonds ويحطمها

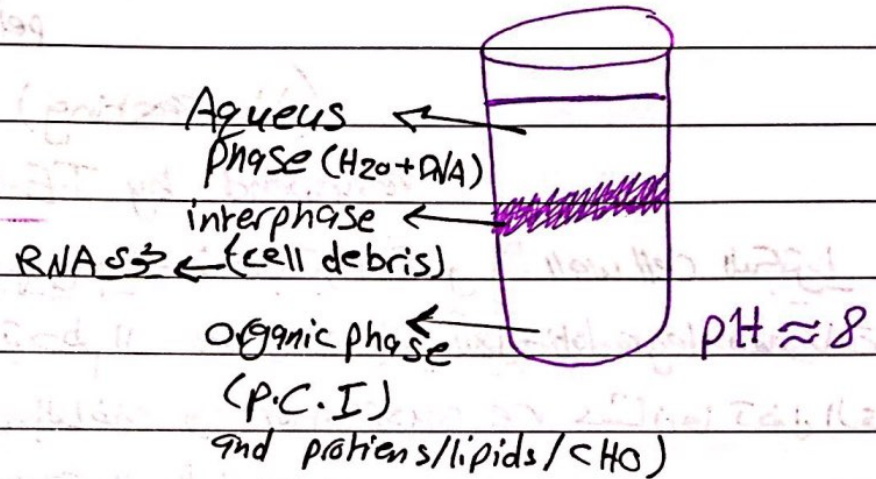


خطوة ٦ : equal volume of phenol / chloroform / isoamyl alcohol : 25 : 24 : 1

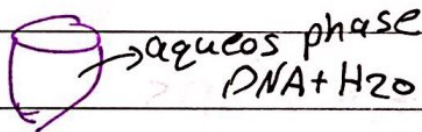
Organic solvents
هنا هو المذيبات

خطوة ٧ :

* بعد ال centrifuge : ال chloroform اتقل الشئ فيهم بالتركيز لما نخطو بنزل كلسي صوة لتحتو وسجل تدخلهم



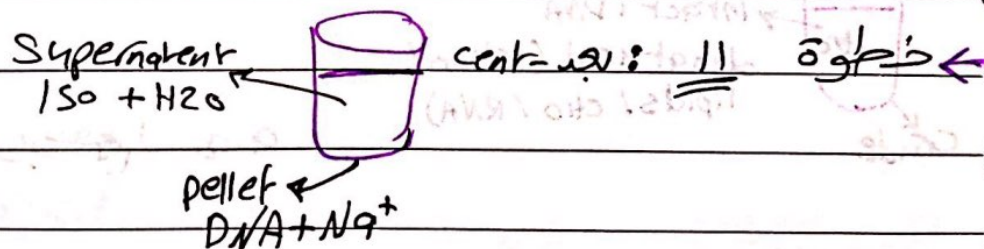
خطوة ٨ : لسحب الطبقه اللي فوق يلي فيها ال DNA وخطها ب new tube



خطوة ٩ : DNA precipitation :

خطوة ١٠ : by Sodium acetate & Isopropanol

(شرفناهم قبل)



خطوة ١٢ : DNA washing :

by 70% ethanol (شرفناهم)

خطوة ١٣ : pipette aspirate

↓ تتابع شرح نفس الخطوات

۱۳

۱۲ نیجی نتخلص من ال Supernat- فابیسیم (discard) عشان

نیون ال DNA بال ethanol ماسوینهد ؟

بیسیم بال pipette Aspirate

و مینهد air dry

Rehydration : ۱۴ 

(TE) buffer

* DNA isolation from blood cell

using salting out
(Promega Kit)

Fresh blood

RBCs lysis

WBCs lysis

Protein precipitation

DNA precipitation

DNA washing

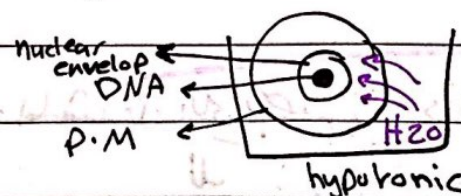
DNA rehydration

* EDTA ترتبط مع Ca^{+}

يمنع ال coagulation لـ Ca^{+} واليونات
الكلكتين Ca^{+} على ال

* methods of isolation: Differential Solubility:

1 cell lysis solution $\rightarrow$ Hypotonic solution (TE + Salts)



2 Nuclei lysis $\rightarrow$ Hypotonic solution (TE + Salts)

هون نفس اللي فوق بس ال Conc. اختلف

تابع $\rightarrow$

450

Nuclei lysis

- Salt: stabilise DNA as double helix.
- Detergent: disrupts lipid layers / help to dissolve membrane / binds positive charges of histone to release DNA
- Elevated temp 80°C : denature proteins and inhibit DNA degradation by DNases

[3] protein precipitation → Salt: precipitate protein & lipids (Hypertonic)

~~DNA precipitation - Sodium salt~~

البروتين hydration shell breakdown by Salt
وال DNA ويتسحب معها البروتين تحت (interaction)

DNA + H_2O ← Centrifuge
pellet

→ DNA precipitation

ونفس نفس خطوات ال e.coli

* الحاشية كل التي تبصر (نشاء ال isolation)

ال DNA كله

- Shearing DNA / DNases / Chemicals

Lab 3

* RNA isolation from blood cells
Using adsorption column (Spin)
↳ silica column

* Types of RNA: All of them is -ve charge

1) mRNA → transfer DNA code to ribosome for translation (2.5-5%)

2) tRNA → brings a.a to ribosomes for protein synthesis

3) rRNA → in ribosomes (highest percent of RNA 90%)

* From mRNA sequence we can estimate a.a seq for protein.

* From mRNA length we can estimate alternative splicing

∴ some information can't get from DNA / only from RNA level

* inactivation RNase enzymes:

All solution used during RNA solution must be

treated with 1% DEPC (Diethyl pyrocarbonate)

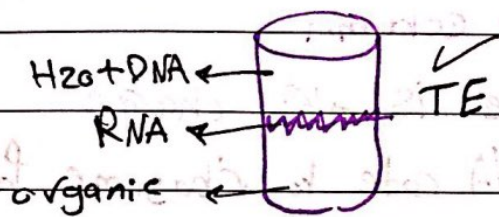
that modify Histidine then inactive RNase

* RNase is active with Mg^{+2} as a cofactor. To inactivate RNase, we use

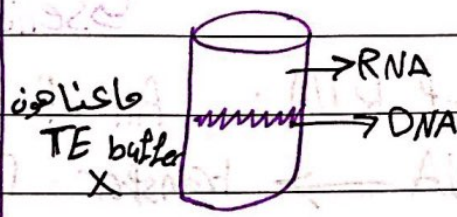
EDTA which chelates Mg^{+2} and 1% DEPC which modifies the active site of RNase.

in DNA isolation

RNA isolation



p.c.I
pH ~ 8



RNA isolation ~ 1% DEPC

Acidic p.c.I
pH ~ 4

RNAase active pH ~ 8
denatured RNA active denaturation optimum for RNA
pH ~ 4
interphase

* Isolation methods for RNA :

- Organic Solvents

- Adsorption methods

- Density gradient centrifugation

1 Cell lysis → Guanidine thiocyanate (chaotropic agent)

- denature proteins / solubilize lipid

membrane / inactivate RNases

denaturation / breakdown :
proteins / lipids for H-bond / Hydrophobic bond

H₂O / CH₃OH / DNA Vander Waals bond

→ B-mercaptoethanol : denatured protein

[2] RNA adsorption:

Guanidine thiocyanate:

dehydrate RNA forcing it to bind to the silica membrane.

Ethanol: Favours the binding of RNA into silica but not the DNA.

Salt: act as cation bridge between the silica & the RNA.

[3] Washing of RNA

نفسهم ال ٣ اللى فوق

[4] RNA elution → RNase-free water: removes chaotropic salt and sodium salt.

* Steps for procedure: RBCs & WBCs for pelleting cells

خطوة ١: 750M لان هون ال DNA اكثر من التجارب اللى قبل فنزله الحبة عشان ال RNA تنزل نسبتهم.

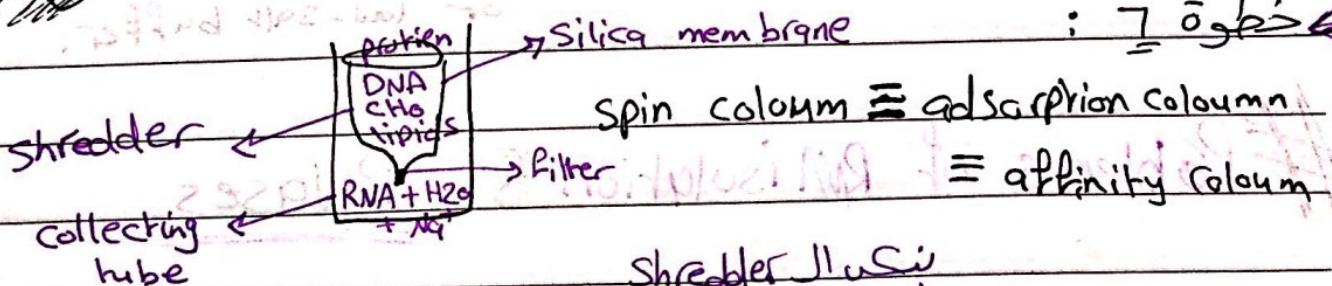
خطوة ٢: E1 (Erythrocytes lysis buffer)

[degradation of RBC'S] → RBC'S burst

خطوة ٣: RLT buffer ≡ lysis buffer

هو اللى فيه Sodium salt / 2-mercapto / Guanidine

خطوة ٤: 7



نكسب ال shredder
ونطير ال collecting